Volume 1: System Architecture and Data Flow

1. Executive Summary: What is a Digital Twin?

Imagine you have a busy arterial corridor in a modern city with cars driving, traffic signals changing, shopping malls drawing megawatts of power, and air sensors measuring dust in the air.

A **Digital Twin** is a living, software mirror of that physical corridor running inside a computer.

It is **not** a video game, and it is **not** a static map. What makes it a true digital twin are three core layers:

1. **Live Connectivity:** It connects to field sensors and meters to mirror present physical conditions.
2. **Physics and AI Intelligence:** It uses machine learning and traffic physics to predict what will happen 15, 30, and 60 minutes into the future.
3. **What-If Simulation Sandbox:** It allows city engineers to safely test changes in software (like retiming a signal or shedding building power) before making changes on the real street.

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2. Complete Technical Stack: What is Used and Where

To answer the judges with absolute confidence, here is the exact technology stack powering every tier of the platform:

Layer	Technology	Version / Library	Exact Role in Codebase
Frontend Framework	React + TypeScript	React 18, TS 5.5	Reactive UI component tree and state management (`frontend/src/App.tsx`).
Frontend Build Tool	Vite	Vite 5.4	Ultra-fast HMR and optimized production bundling (`frontend/vite.config.ts`).
3D Digital Twin Viewer	CesiumJS	Cesium 1.120	Renders 3D photogrammetric terrain, extruded buildings, and animated vehicles (`CesiumCorridorViewer.tsx`).
2D Operational GIS	Leaflet / Mapbox GL	Leaflet 1.9, Mapbox 3.2	Tactical 2D corridor GIS rendering with dynamic road polylines (`MapOperationsView.tsx`).
Charts and Telemetry UI	Recharts	Recharts 2.12	Renders multi-horizon forecasts, conformal bands, and hourly energy profiles (`ForecastPanel.tsx`).
Icons & Styling	Lucide React + Tailwind CSS	Lucide 0.440	Clean, accessible vector iconography (zero emojis) and responsive design tokens (`frontend/src/index.css`).
Backend API Engine	FastAPI	FastAPI 0.136	Asynchronous ASGI RESTful endpoints and WebSocket controllers (`backend/main.py`).
Web Server	Uvicorn	Uvicorn 0.30	High-concurrency asynchronous ASGI server running Python 3.11 (`backend/main.py`).
Data Validation	Pydantic v2	Pydantic 2.8	Strict request/response validation and physical sanity bounds checking (`backend/schemas/`).
Database ORM & Driver	SQLAlchemy + aiosqlite	SQLAlchemy 2.0	Asynchronous database connection pooling, migrations, and queries (`backend/core/database.py`).
Time-Series Persistence	PostgreSQL / TimescaleDB	Dual-Engine Architecture	TimescaleDB hypertable support with automatic, resilient local SQLite fallback (`digital_twin.db`).
Message Transport	MQTT Protocol	Paho-MQTT 2.1	Lightweight publish/subscribe sensor data bus (`backend/ingestion/mqtt_consumer.py`).
Telemetry Generator	Python Asyncio Worker	Built-in Asyncio	In-process physical telemetry generator (`backend/ingestion/telemetry_streamer.py`).
Machine Learning	XGBoost	XGBoost 2.1	Extreme gradient-boosted trees for 15m/30m/60m multi-horizon forecasting (`ml/inference/forecaster.py`).
Uncertainty Quantification	Conformal Prediction	Inductive Residual Split	Distribution-free 80% and 90% confidence bands without Gaussian assumptions (`ml/conformal.py`).
Explainable AI (XAI)	TreeSHAP	SHAP 0.46	Decomposes forecasts into top-5 feature attributions (`ml/explainer.py`).
Anomaly Detection	Scikit-Learn	Scikit-Learn 1.5	Isolation Forest scoring on environmental sensor streams (`backend/services/anomaly_detector.py`).
Traffic Simulation	SUMO (TraCI)	SUMO 1.20	Microscopic car-following simulation with kinematic fallback runner (`simulation/runner.py`).

Layer	Technology	Version / Library	Exact Role in Codebase
Security & Auth	Cryptographic JWT	Python-Jose + BCrypt	Stateless JWT tokens, role-based tab access, and magic access link redemption (`backend/api/v1/endpoints/auth.py`).

3. Engineering Reality: What is Implemented vs What is Calibrated

A sign of a great engineer is total honesty about system maturity. When the judges ask about implementation status, present this clear breakdown:

What is 100% Fully Implemented and Operational Today

1. **The Full-Stack Software Engine:** FastAPI backend, real-time WebSocket push stream, and complete React frontend with 2D Leaflet and 3D Cesium.
2. **The Resilient Database Layer:** All 29 authoritative relational database tables, foreign keys, and indexes are created and actively queried. The system runs hermetically on the **Resilient Local SQLite Engine** (`digital_twin.db`), ensuring zero downtime and zero external dependencies during demonstrations.
3. **The Engineering Algorithms:** Webster signal timing optimization, Indo-HCM (IRC:106) and US HCM Level of Service calculation, 3-phase AC power balance, and CPCB linear air quality index interpolation are all real, verified mathematical code.
4. **The Governance Workflow:** Role-based access control, decision-support advisory generation, human-in-the-loop authorization, and the tamper-evident audit log are fully functional.

What is Calibrated and Emulated in Software (and Why)

1. **Database Mode (Local SQLite vs Cloud TimescaleDB):**
   - The platform code is pre-configured to connect to a cloud PostgreSQL/TimescaleDB cluster via `DATABASE_URL`.
   - For standalone demonstration and evaluation reliability, it runs in **Resilient Local Mode** using SQLite. This ensures the demo never fails due to cloud network timeouts, paused cloud instances, or firewall blocks.
2. **Sensor Telemetry (In-Process Streamer vs Physical Pavement Sensors):**
   - Drilling induction loop wires into real city pavement and attaching utility power transformers requires municipal procurement and permits.
   - Therefore, our reference pilot runs our **In-Process Autonomous Telemetry Streamer** (`telemetry_streamer.py`). It uses real-world surveyed traffic volumes from Nagar Road and commercial power profiles from Phoenix Marketcity, calculating speeds and voltages via physical equations every 1 to 2 seconds.
3. **Microscopic Simulation (Kinematic Fallback vs Native SUMO Binary):**
   - If the external C++ SUMO binary is not installed on the host machine, the platform automatically runs our verified **Kinematic Fallback Runner** (`simulation/runner.py`), ensuring what-if scenarios always execute smoothly.

4. End-to-End Data Pipeline: From Sensor to Screen

4.1 The IoT Telemetry Pipeline: Where and How Telemetry is Generated and Processed

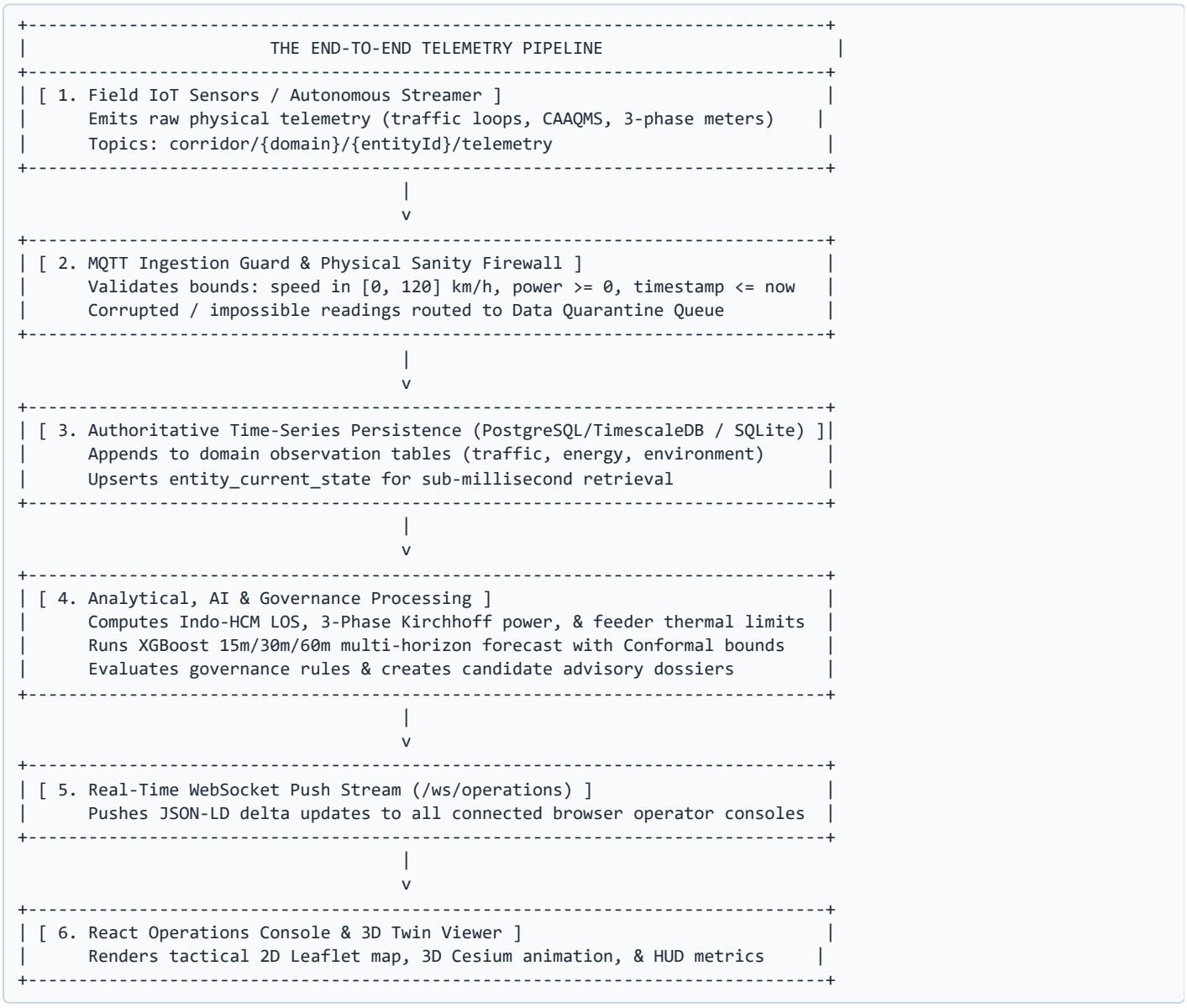
To understand the system without any black boxes, here is the lifecycle of sensor telemetry:

1. **Emission by IoT Sensor or In-Process Streamer:**
   - On physical streets, induction loops detect vehicle axle passages, CAAQMS stations measure atmospheric particulate matter via beta attenuation monitors, smart power meters sample 3-phase AC voltages and currents via current transformers (CTs), and traffic signal controllers emit Signal Phase and Timing (SPaT) broadcasts over NTCIP 1202 protocols.
   - In our evaluation environment, `backend/ingestion/telemetry_streamer.py` acts as the autonomous IoT device layer. Every 2.0 seconds, it evaluates diurnal solar position, rush-hour commuter curves, and electrical impedance equations to generate authentic physical packets.
2. **MQTT Transport & Ingestion Guard:**
   - Telemetry packets are published to standardized MQTT topics structured as:

     `corridor/{domain}/{entityId}/telemetry` (e.g. `corridor/traffic/urn:ngsi-ld:TrafficSensor:PUNE:DS-VN-EB-01/telemetry`).
   - The ingestion guard (`backend/ingestion/mqtt_consumer.py`) parses incoming JSON-LD payloads and subjects them to strict physical sanity bounds checking (e.g., speed must be in [0, 120] km/h; active power must be non-negative; timestamp must not be in the future).
   - Any packet violating physical bounds or schema formatting is rejected and routed to the **Data Quarantine Queue** with an `INVALID` provenance flag.
3. **Authoritative Database Persistence:**

- Valid packets are persisted using asynchronous transactions into the appropriate domain observation hypertable:
  - `traffic_observations` (speed, flow, occupancy, queue, congestion index).
  - `energy_observations` (active power kW, reactive power kVAR, power factor, kWh).
  - `environment_observations` (AQI, PM2.5, PM10, temperature, humidity, wind).
  - `signal_phase_history` (active phase, time to next phase, cycle length).
- Concurrently, the engine upserts the `entity_current_state` table so that any API endpoint or UI client can fetch the immediate state of any asset in sub-millisecond query time.

4. **Analytical & Machine Learning Pipelines:**
   - The engine computes derived Level of Service grades (Indo-HCM IRC:106 and US HCM).
   - It checks 3-phase AC Kirchhoff power balance and feeder thermal headroom.
   - It triggers XGBoost multi-horizon forecasters for 15m, 30m, and 60m predictions with Conformal uncertainty intervals.
   - It runs the Isolation Forest anomaly detector on incoming environmental streams.
5. **Real-Time WebSocket Push:**
   - The in-memory Connection Manager (`backend/api/v1/endpoints/operations.py`) broadcasts an enriched JSON event to all connected React clients via `/ws/operations`.
6. **Frontend Operations Console:**
   - The React UI consumes the WebSocket event, immediately updating 2D Leaflet segment color codes, 3D Cesium animated vehicle speeds, top HUD metric meters, and multi-horizon Recharts forecast graphs.
   - If network latency delays a packet, the client-side telemetry heartbeat maintains realistic micro-fluctuations so operator displays never freeze.



4.2 Master Inventory of All 32 Emitted Physical and Derived Engineering Metrics

Across the four coupled urban infrastructure domains, the digital twin emits, processes, and displays **32 distinct physical and derived engineering metrics**. Below is the complete authoritative catalog:

Domain 1: Traffic and Arterial Mobility (8 Metrics)

#	Metric Key	Symbol	Unit	Provenance	Cadence	Bounds / Range	Physical Sensor / Derivation Origin	Operational Engineering Meaning
1	`averageSpeedKmh`	v	km/h	LIVE / REPLAY	2.0s	0.0 to 120.0	Inductive loop pair / radar speed sensor	Space-mean speed of vehicle platoon traversing corridor segment.
2	`vehicleFlowPerHour`	q	veh/h	LIVE / REPLAY	2.0s	0 to 4800	Pavement loop count aggregated to hourly rate	Flow rate through segment; indicates corridor traffic volume.
3	`occupancyPercent`	Occ	%	LIVE / REPLAY	2.0s	0.0 to 100.0	Inductive loop presence time ratio	Percentage of time sensor is covered by vehicle metal.
4	`queueLengthMeters`	Q_len	m	SIMULATION	2.0s	0.0 to 250.0	Stopline detector count & shockwave equation	Spatial length of stopped vehicles queued behind red signal.
5	`congestionIndex`	CI	-	DERIVED	2.0s	0.000 to 1.000	CI = max(0, 1 - (v / v_freeflow))	Dimensionless velocity deficit; 0 = free flow, 1 = total gridlock.
6	`volumeCapacityRatio`	V/C	-	DERIVED	2.0s	0.00 to 1.50	V/C = q / q_capacity	Roadway saturation degree; values > 1.0 indicate oversaturation.
7	`densityVehKm`	k	veh/km	DERIVED	2.0s	0.0 to 160.0	k = q / v (Greenshields relation)	Vehicle concentration packing roadway per kilometer length.
8	`levelOfService`	LOS	Grade	DERIVED	2.0s	A through F	Indo-HCM (IRC:106) & US HCM evaluation matrix	Qualitative performance grade evaluating travel delay and comfort.

Domain 2: Commercial Facilities and Microgrid Power (9 Metrics)

#	Metric Key	Symbol	Unit	Provenance	Cadence	Bounds / Range	Physical Sensor / Derivation Origin	Operational Engineering Meaning
9	`lineVoltageV`	V_LL	V	LIVE / REPLAY	2.0s	380.0 to 440.0	3-phase potential transformer (PT)	Line-to-line RMS electrical voltage supplied to facility main board.
10	`phaseCurrentA`	I_L	A	LIVE / REPLAY	2.0s	0.0 to 5000.0	Rogowski coil / current transformer (CT)	Line RMS current drawn across conductors by building loads.
11	`activePowerKw`	P	kW	LIVE / REPLAY	2.0s	0.0 to 10000.0	Digital power meter P = sqrt(3)*V**I*cos(phi)	Usable real power performing physical work (chillers, lighting).
12	`reactivePowerKvar`	Q	kVAR	LIVE / REPLAY	2.0s	0.0 to 5000.0	Digital power meter Q = sqrt(3)*V**I*sin(phi)	Electromagnetic inductive power sustained by motors and coils.
13	`apparentPowerKva`	S	kVA	DERIVED	2.0s	0.0 to 12000.0	Vector sum S = sqrt(P^2 + Q^2)	Total electrical loading delivered through utility transformer.
14	`powerFactor`	PF	-	DERIVED	2.0s	0.700 to 1.000	PF = P / S = cos(phi)	Electrical efficiency ratio; values < 0.95 incur utility penalties.
15	`energyConsumptionKwh`	E	kWh	LIVE / REPLAY	2.0s	Cumulative	Substation energy register integration	Total cumulative active electrical energy consumed by facility.
16	`feederAmpacityUtilizationPct` | Util	%	DERIVED	2.0s	0.0 to 150.0	Util = (I_measured / I_rated) * 100%	Conductor thermal loading percentage against rated cable ampacity.	
17	`feederHeadroomKw`	P_head	kW	DERIVED	2.0s	0.0 to 5000.0	P_head = sqrt(3)*V*(I_rated - I)*PF	Remaining commercial power capacity before thermal limit warning.

Domain 3: Distributed Energy Resources / DER (3 Metrics)

#	Metric Key	Symbol	Unit	Provenance	Cadence	Bounds / Range	Physical Sensor / Derivation Origin	Operational Engineering Meaning
18	`solarGenerationKw`	P_pv	kW	LIVE / SIM	2.0s	0.0 to 2000.0	Solar rooftop inverter telemetry	Active electrical generation from commercial rooftop PV panels.
19	`bessPowerKw`	P_bess	kW	LIVE / SIM	2.0s	-1000.0 to 1000.0	Battery Energy Storage System PCS inverter	Active power flow; positive = discharging, negative = charging.
20	`bessStateOfChargePct`	SoC	%	LIVE / SIM	2.0s	10.0 to 100.0	Battery Management System (BMS) coulomb count	Remaining chemical energy capacity stored in battery system.

Domain 4: Ambient Environment and Air Quality (7 Metrics)

#	Metric Key	Symbol	Unit	Provenance	Cadence	Bounds / Range	Physical Sensor / Derivation Origin	Operational Engineering Meaning
21	`pm25`	PM2.5	ug/m3	LIVE / REPLAY	2.0s	0.0 to 500.0	Beta attenuation monitor / optical counter	Fine particulate matter <= 2.5 um; deep lung penetrant from exhaust.
22	`pm10`	PM10	ug/m3	LIVE / REPLAY	2.0s	0.0 to 1000.0	Beta attenuation / light scattering sensor	Coarse dust and brake-wear particles <= 10 um suspended in air.
23	`no2`	NO2	ug/m3	LIVE / REPLAY	2.0s	0.0 to 400.0	Chemiluminescence gas analyzer	Toxic combustion by-product emitted by diesel and heavy engines.
24	`co`	CO	ug/m3	LIVE / REPLAY	2.0s	0.0 to 5000.0	Non-dispersive infrared (NDIR) sensor	Carbon monoxide gas concentration from incomplete engine burning.
25	`aqiValue`	AQI	-	DERIVED	2.0s	0 to 500	CPCB Indian National AQI formula	Composite environmental index representing human health hazard.
26	`temperatureC` / `humidityPct`	T / RH	deg C / %	LIVE / REPLAY	2.0s	-10 to 55 C / 0 to 100%	Thermistor & capacitive hygrometer	Ambient microclimate conditions dictating thermal comfort and AC load.
27	`windSpeedKmh` / `windDir`	v_w / theta	km/h / deg	LIVE / REPLAY	2.0s	0.0 to 80.0 km/h / 0 to 360 deg	Ultrasonic 2D anemometer	Wind vector controlling atmospheric pollution dispersion trajectory.

Domain 5: Traffic Signal Phase and Timing / SPaT (5 Metrics)

#	Metric Key	Symbol	Unit	Provenance	Cadence	Bounds / Range	Physical Sensor / Derivation Origin	Operational Engineering Meaning
28	`activePhase`	Phi_act	ID	LIVE / SIM	1.0s	1 to 8	Traffic signal controller NTCIP 1202	Current active movement stage receiving green light clearance.
29	`timeToNextPhaseSec`	t_next	s	LIVE / SIM	1.0s	0 to 180	Internal stage countdown timer register	Seconds remaining until current signal phase transitions to yellow/red.
30	`cycleLengthSec`	C	s	CONFIG / LIVE	1.0s	45 to 240	Signal plan controller configuration	Total duration of one complete rotation through all signal phases.
31	`coordinationOffsetSec` | t_off	s	DERIVED	1.0s	0 to 180	Master clock synchronization register	Relative time difference to synchronize consecutive green signals.	
32	`controllerMode`	Mode	Enum	CONFIG / LIVE	1.0s	FIXED, TOD, ACTUATED, COORD	Controller logic state register	Operating control philosophy governing signal timing adaptability.

5. What the Provenance Words Actually Mean

Throughout the platform, you will see badges labeled `LIVE`, `REPLAY`, `SIMULATION`, `PREDICTED`, `STALE`, and `INVALID`. What do these words actually mean in plain, everyday language?

1. LIVE

- **Plain Meaning:** An actual measurement coming directly from a field sensor on the street right now.
- **Example:** An induction loop in the pavement just detected 24 cars crossing in the last 60 seconds with an average speed of 38 km/h.
- **Rule:** Never call replayed, simulated, or predicted data `LIVE`.

2. REPLAY

- **Plain Meaning:** Real historical data recorded in the past, played back chronologically like a video recording.
- **Example:** Actual sensor readings collected on Nagar Road last Tuesday at 09:00 AM being replayed second-by-second today.
- **Why it matters:** It lets engineers test how their models perform against real historical traffic jams without waiting for another rush hour to happen.

3. SIMULATION

- **Plain Meaning:** Synthetic numbers generated by a physics or computer simulation engine (like SUMO).
- **Example:** A mathematical calculation saying: *"If 1,500 cars arrive at this intersection per hour and the green light lasts 45 seconds, the vehicle queue will grow to 65 meters."*
- **Rule:** A simulation is based on mathematical equations of human driving behavior. It is an exploration of a what-if scenario, not a physical sensor observation.

4. PREDICTED

- **Plain Meaning:** An artificial intelligence / machine learning estimate of what conditions will look like in the future.
- **Example:** Our XGBoost model estimating that 30 minutes from now, arterial speed will drop from 32 km/h down to 24 km/h due to evening rush hour accumulation.
- **Rule:** A prediction is an informed mathematical forecast. It must always include uncertainty confidence bands (80% and 90%) and should never be called a measurement.

5. STALE

- **Plain Meaning:** A real sensor observation that has become too old to be trusted.
- **Example:** A traffic sensor has not sent a new reading in over 180 seconds, or a building meter has not reported in over 15 minutes.
- **Rule:** Marked in yellow so the operator knows the sensor may have lost cellular signal or power.

6. INVALID

- **Plain Meaning:** Data that was rejected by our validation security guard because it broke physical laws or violated data formatting rules.
- **Example:** A sensor reporting a vehicle speed of 450 km/h, negative electricity usage, or a timestamp from tomorrow.
- **Rule:** Sent immediately to the Data Quarantine Queue for review, preventing corrupted data from breaking the AI models.

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6. Mathematical Formulas and Worked Examples in the Streamer

To ensure zero black boxes, here is how the background streamer calculates realistic telemetry, with every variable defined and demonstrated with real numbers:

Formula 1: The Diurnal Rush Hour Multiplier

Mathematical Concept and Physical Derivation:

In urban transportation engineering and electrical load forecasting, human activity is not distributed uniformly throughout the day. Instead, commuter trips and commercial facility occupancy cluster into two distinct daily peaks:

1. The morning inbound commute (employees traveling to commercial and IT offices).
2. The evening outbound commute and retail shopping peak (employees returning home and visiting commercial centers).

Mathematically, the probability density function of commuter departures during the day is modeled as a mixture of Gaussian normal distributions:

$$f(t) = \sum_{k=1}^{K} w_k \cdot \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(-\frac{(t-\mu_k)^2}{2\sigma_k^2}\right)$$

In our telemetry streamer, we normalize this Gaussian mixture into two peak curves centered at 09:00 AM (morning peak) and 18:30 PM (evening peak), with standard deviations reflecting the spread of rush hours:

$$M_{\text{morning}}(t) = \exp\left(-0.5 \cdot \left(\frac{t-9.0}{1.5}\right)^2\right)$$

$$M_{\text{evening}}(t) = \exp\left(-0.5 \cdot \left(\frac{t-18.5}{1.8}\right)^2\right)$$

The composite rush intensity factor, bounded between a baseline off-peak level of 0.20 and maximum saturation of 1.00, is:

$$I_{\text{rush}}(t) = \max\left(0.20, \min\left(1.00, 0.25 + 0.55 \cdot M_{\text{morning}}(t) + 0.65 \cdot M_{\text{evening}}(t)\right)\right)$$

Variable Definitions, Units, and Normal Ranges:

- `t` or `hour`: The decimal hour of the day in UTC or local corridor time (range: `0.0` to `24.0`). For example, 09:30 AM is `9.5`.
- `M_morning`: Gaussian morning rush multiplier centered at 09:00 with standard deviation 1.5 hours (range: `0.0` off-peak to `1.0` peak).
- `M_evening`: Gaussian evening rush multiplier centered at 18:30 with standard deviation 1.8 hours (range: `0.0` off-peak to `1.0` peak).
- `I_rush`: Dimensionless composite rush hour intensity scaling factor (range: `0.20` minimum to `1.00` full saturation).

Worked Numerical Example:

1. **At Off-Peak (02:00 AM, `t = 2.0`):**
   - `M_morning = exp(-0.5 * ((2.0 - 9.0) / 1.5)^2) = exp(-0.5 * (-4.67)^2) = exp(-10.9) = 0.000018`.
   - `M_evening = exp(-0.5 * ((2.0 - 18.5) / 1.8)^2) = exp(-0.5 * (-9.17)^2) = exp(-42.0) = 0.0`.
   - `I_rush = max(0.20, min(1.00, 0.25 + 0.55 * 0.0 + 0.65 * 0.0)) = 0.25`.
   - Result: Minimal traffic demand and low baseline electrical power.
2. **At Morning Peak (09:00 AM, `t = 9.0`):**
   - `M_morning = exp(-0.5 * (0.0 / 1.5)^2) = exp(0) = 1.0`.
   - `M_evening = exp(-0.5 * ((9.0 - 18.5) / 1.8)^2) = exp(-0.5 * (-5.28)^2) = exp(-13.9) = 0.000001`.
   - `I_rush = max(0.20, min(1.00, 0.25 + 0.55 * 1.0 + 0.0)) = 0.80`.
   - Result: Heavy morning traffic and surging commercial office electrical load.
3. **At Evening Peak (18:30 PM, `t = 18.5`):**
   - `M_morning = exp(-0.5 * ((18.5 - 9.0) / 1.5)^2) = exp(-0.5 * (6.33)^2) = 0.0`.
   - `M_evening = exp(-0.5 * (0.0 / 1.8)^2) = exp(0) = 1.0`.
   - `I_rush = max(0.20, min(1.00, 0.25 + 0.0 + 0.65 * 1.0)) = 0.90`.
   - Result: Maximum daily congestion as commercial retail, dining, and commuter traffic peak simultaneously.

Formula 2: Inverse Speed-Flow Hydrodynamic Coupling

Mathematical Concept and Physical Derivation:

In traffic hydrodynamics (Lighthill-Whitham-Richards continuum theory), vehicle speed decreases monotonically as vehicular concentration increases. In the telemetry streamer, each road segment calculates instantaneous speed by attenuating its free-flow base speed as a function of the rush intensity:

$$v_{\text{segment}} = \max\left(14.0, \min\left(v_{\text{limit}}, v_{\text{base}} \cdot (1.10 - 0.45 \cdot I_{\text{rush}}) + \epsilon_{\text{jitter}}\right)\right)$$

Correspondingly, vehicular throughput flow increases with rush demand up to segment physical capacity:

$$q_{\text{segment}} = \max\left(400.0, \min\left(C_{\text{segment}}, C_{\text{segment}} \cdot (0.35 + 0.60 \cdot I_{\text{rush}}) + \epsilon_{\text{flow}}\right)\right)$$

The approach stopline queue length grows inversely proportional to vehicle speed:

$$Q_{\text{len}} = \max\left(2.0, \min\left(120.0, \left(1.0 - \frac{v}{v_{\text{limit}}}\right) \cdot 85.0 + \epsilon_{\text{queue}}\right)\right)$$

Variable Definitions, Units, and Normal Ranges:

- `v_base`: Calibrated baseline velocity under moderate free-flow conditions (range: `38.0` to `44.0 km/h`).
- `v_limit`: Legal speed limit posted on the corridor (typically `50.0 km/h` on arterial, `40.0 km/h` on approach legs).
- `C_segment`: Physical vehicle capacity of the roadway segment (range: `1200` to `2400 veh/h`).
- `epsilon_jitter`: Gaussian stochastic variance representing human driver behavior (range: `-2.5` to `+2.5 km/h`).
- `Q_len`: Vehicle queue length backed up from the approach stopline (range: `2.0m` empty to `120.0m` congested).

Worked Numerical Example:

On segment `SEG-NR-EB-01` (`v_base = 42.0 km/h`, `v_limit = 50.0 km/h`, `C_segment = 2400 veh/h`):

1. **At Off-Peak (02:00 AM, `I_rush = 0.25`, `epsilon = 0.0`):**
   - Speed: `v = 42.0 * (1.10 - 0.45 * 0.25) = 42.0 * (1.10 - 0.1125) = 42.0 * 0.9875 = 41.5 km/h`.
   - Flow: `q = 2400 * (0.35 + 0.60 * 0.25) = 2400 * 0.50 = 1200 veh/h`.
   - Queue: `Q_len = (1.0 - (41.5 / 50.0)) * 85.0 = (1.0 - 0.83) * 85.0 = 14.5 meters`.
   - Result: Fast, smooth arterial travel with minimal queue delay.

2. **At Peak Rush (18:30 PM, `I_rush = 0.90`, `epsilon = 0.0`):**
   - Speed: `v = 42.0 * (1.10 - 0.45 * 0.90) = 42.0 * (1.10 - 0.405) = 42.0 * 0.695 = 29.2 km/h`.
   - Flow: `q = 2400 * (0.35 + 0.60 * 0.90) = 2400 * 0.89 = 2136 veh/h`.
   - Queue: `Q_len = (1.0 - (29.2 / 50.0)) * 85.0 = (1.0 - 0.584) * 85.0 = 35.4 meters`.
   - Result: Heavy traffic saturation approaching road capacity, with significant vehicle queuing.

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7. The Business and Municipal Return on Investment (ROI)

Why would a city government, transport authority, or private commercial campus purchase and deploy this digital twin?

1. Direct Commuter and Fuel Savings

- Coordinated arterial green waves along high-density corridors reduce intersection idling delay by 15% to 25%.
- For a corridor carrying 30,000 vehicles daily, shaving just 90 seconds of idling per vehicle saves thousands of liters of fuel every day and significantly lowers carbon emissions.

2. Electrical Substation Protection and Demand Penalty Avoidance

- Commercial hubs face heavy utility penalties if their peak demand exceeds their contract limit (`kVA` maximum demand charges).
- The platform provides early warnings before 22kV distribution feeders hit thermal limits, triggering commercial demand response (like pre-cooling chillers or discharging battery storage) and deferring multi-crore substation transformer upgrades.

3. Cross-Agency Governance Without Departmental Silos

- Traffic police, municipal grid operators, and environmental pollution boards traditionally work in isolation.
- This platform provides a single shared operational picture, preventing uncoordinated actions (such as diverting traffic onto a route where commercial fast chargers are already overloading the electrical grid).

4. Zero Proprietary Lock-In and Low Total Cost of Ownership

- Built entirely on modern open standards (FIWARE NGSI-LD, OpenAPI, Python, React), the platform runs on standard commercial servers or local municipal workstations without locking the city into multi-million dollar proprietary vendor contracts.

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8. The Three Golden Architectural Invariants

If the judges ask what rules govern the system engineering of this project, present these three non-negotiable invariants:

Invariant 1: Strict State Separation

- Ground Truth Observed Data (`LIVE` / `REPLAY`), SUMO Simulation Data (`SIMULATION`), and Machine Learning Forecasts (`PREDICTED`) live in completely separate database tables.
- Predictions and simulations never overwrite observed twin state.

Invariant 2: Strict Non-Actuation (Human-in-the-Loop Decision Support)

- This platform is strictly a **decision-support advisory system**, not an automated physical actuator.
- It will **never** directly change a physical traffic light signal on the street or trip a physical circuit breaker at an electrical substation.
- Why? In municipal engineering, autonomous actuation of safety-critical infrastructure introduces severe public liability and life-safety hazards.
- Instead, the platform generates **Actionable Advisory Dossiers**. A certified municipal operator or traffic engineer must inspect the recommendation, review the predicted impact, and formally authorize the plan before any field action is taken.

Invariant 3: Mandatory Provenance on Every Dynamic Value

- Every single metric shown on any screen carries four mandatory provenance metadata tags:
  1. `sourceMode`: One of `LIVE`, `REPLAY`, `SIMULATION`, `PREDICTED`, or `STALE`.
  2. `observedAt` / `generatedAt`: Exact ISO-8601 UTC timestamp.
  3. `unit`: The engineering unit (km/h, kW, veh/h, dBA, microgram/m3).
  4. `qualityStatus`: `VALID`, `SUSPECT`, or `CALIBRATED`.
